WellBot Global Multilingual Wellness Assistant

AI-powered conversational assistant delivering accessible health information worldwide

Project Introduction

Mission

Deliver accessible, non-diagnostic wellness information based on authoritative sources (WHO/CDC guidelines) through an intelligent, multilingual conversational interface

Core Technologies

Flask

Backend & web framework

JWT Auth

Secure HTTP-only cookies

Rasa

NLU & conversational AI

SQLite

Data persistence layer

WELLBOT
STAY WELL

System Architecture

User Interface

Browser interactions: login, chat requests with JWT validation

Flask Gateway

Authentication, security layer, routes messages to Rasa

Rasa NLU/Core

Intent detection, entity extraction, multilingual processing

Action Server

Database queries, response formatting, returns data

Integrated 3-Component Model

Flask server acts as central gateway, coordinating authentication, conversation flow, and knowledge base access

Secure Data Flow

- JWT-protected endpoints
- Language-aware routing
- Real-time database queries
- Bilingual response delivery

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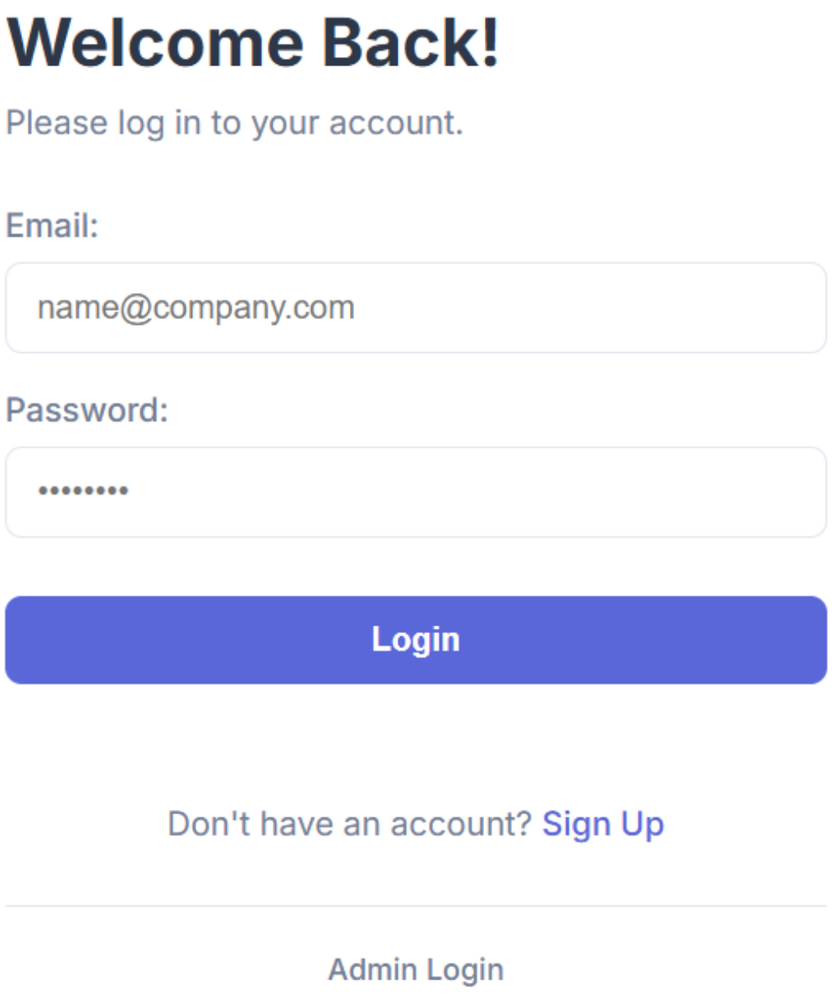

Milestone 1: Foundation

User Authentication & Profile Management

01

User Registration

Secure account creation with salted/hashed passwords, capturing email, age group, preferred language

02

JWT-Based Login

Credential validation with secure HTTP-only cookie session establishment

03

Profile API

Protected endpoints for GET/PUT operations on user preferences and settings

04

Page Security

Automatic redirection for unauthenticated users accessing protected routes

Security First

werkzeug.security for password hashing, JWT tokens in HTTP-only cookies, protected API endpoints

User-Centric Design

Multilingual preference storage, age-appropriate content filtering, persistent session management

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Milestone 2: Core Engine

Conversational AI & Web Interface

Rasa NLU Model

Trained to understand greet, goodbye, ask_symptom, ask_first_aid intents with entity extraction

Knowledge Base

SQLite health_knowledge table storing intent-entity-response mappings for English and Hindi

Custom Actions

ActionQueryKnowledgeBase connects to database, queries based on intent/entity, formats responses

Full-Stack UI

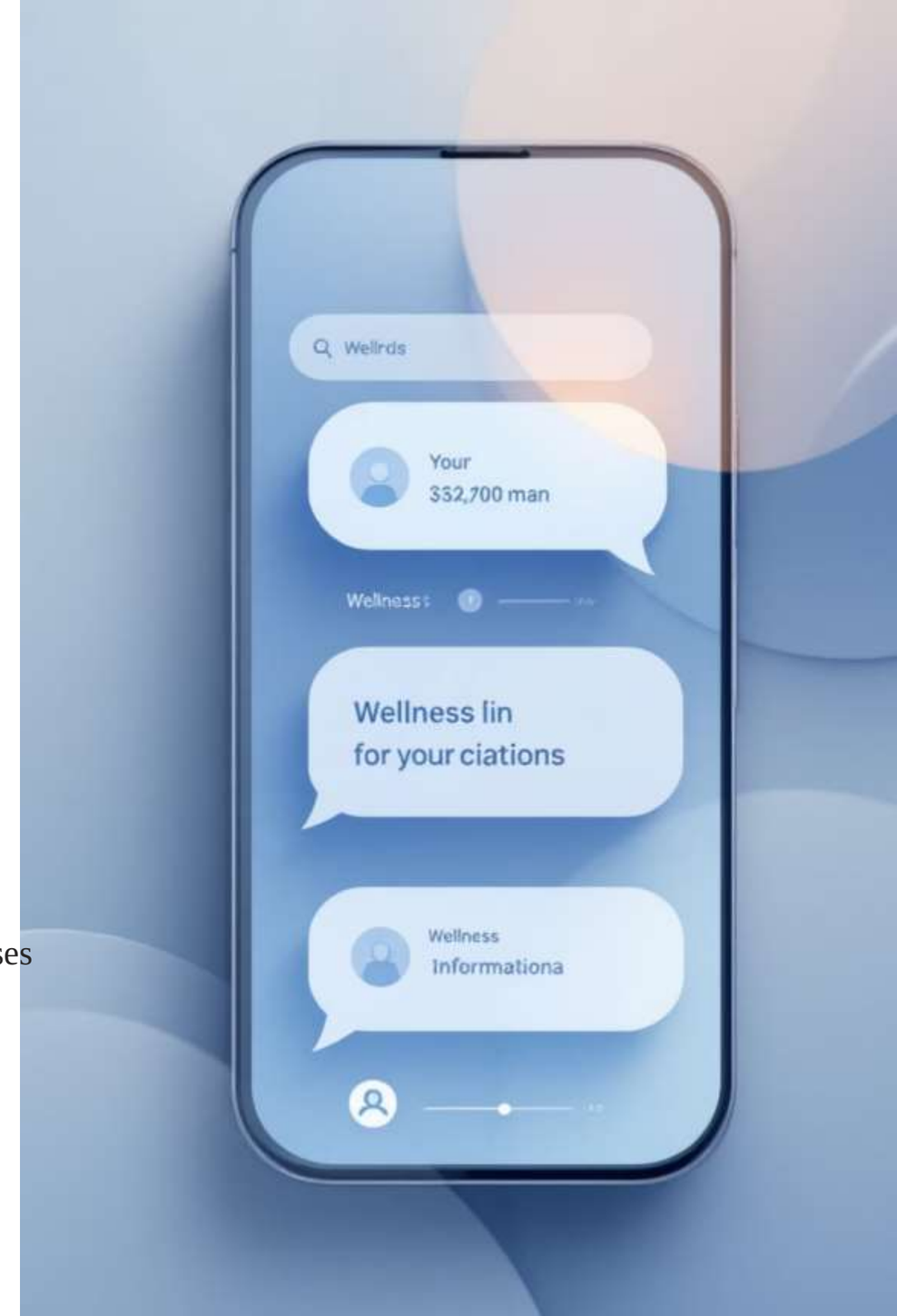
- Flask templates
- Professional CSS
- Dynamic JavaScript

Data Pipeline

- CSV import script
- Pandas processing
- SQLite persistence

API Integration

- /chat endpoint
- Real-time responses
- JSON formatting



Milestone 3: Enhancement

Advanced NLP & Multilingual Expansion

1

100+ Knowledge Entries

Comprehensive health database covering illnesses, first-aid, wellness tips, prevention—all WHO/CDC-styled

2

Ethical Guardrails

Automatic disclaimer at conversation start: "Not a medical professional"

3

Hindi NLU Model

Complete Hindi training dataset (data/hi/nlu.yml) with intent/entity examples

4

Dynamic Language Routing

User preference metadata passed to Rasa, response_en or response_hi selected automatically

5

Advanced Entity Extraction

New entities: duration, severity, location for precise health query understanding

Safety & Ethics

Clear disclaimers, non-diagnostic language, authoritative source alignment

True Bilingual Support

Intent recognition and response generation in both English and Hindi

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Milestone 4: Admin Control

Dashboard & System Refinement

Knowledge Base Management

View, add, delete intent-entity-response mappings directly from web UI

User Analytics

Monitor interaction logs, identify common queries and usage patterns

Feedback Review

Aggregate user ratings and comments to drive continuous improvement

User Feedback Loop

Rating System

Thumbs up/down on every bot response

Comment Box

Qualitative feedback with timestamps

Feedback API

Secure /feedback endpoint with user ID tracking

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UI/UX Excellence

Professional Aesthetic

Consistent indigo/violet theme, modern design language, polished visual hierarchy

Responsive Layout

Two-column chat interface with navigation sidebar, optimized for desktop and mobile

Split-Screen Auth

Enhanced login/registration experience with visual appeal and clear user flow

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Production-Ready

Secure, scalable, multilingual wellness assistant built with best practices

100+

Knowledge Entries

2

Languages

4

Major Milestones